ETF Portfolio Optimization Using Historical Market Data

Monte Carlo Simulation
Mathematical Portfolio Optimization
Practical Portfolio Construction

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Date: 7/22/2026

Why This Project Matters

The business challenge.
Currently, investment managers, financial analysts, and individual investors must decide how to allocate capital across multiple assets while balancing return, risk, and diversification.

Why Portfolio Optimization?

Portfolio optimization uses historical market data and applied quantitative methods to evaluate thousands of potential portfolio combinations. By comparing expected return, volatility, and the Sharpe Ratio, investors can identify allocation strategies that better balance risk and return.

Business Problem:

Investors must determine how to allocate capital across multiple ETFs while balancing expected return and investment risk.

Business Question:

Can historical ETF data be used to construct an optimized portfolio that maximizes risk-adjusted returns while maintaining practical diversification?

Objectives:

- Analyze historical ETF performance
- Compare optimization methods
- Recommend a practical portfolio allocation

Dataset & Methodology

Data Source
Yahoo Finance (yfinance)
Historical Daily ETF Prices

Portfolio Optimization Process

Step 01
Data Collection

Step 02
Data Quality Checks

Step 03
Exploratory Data Analysis

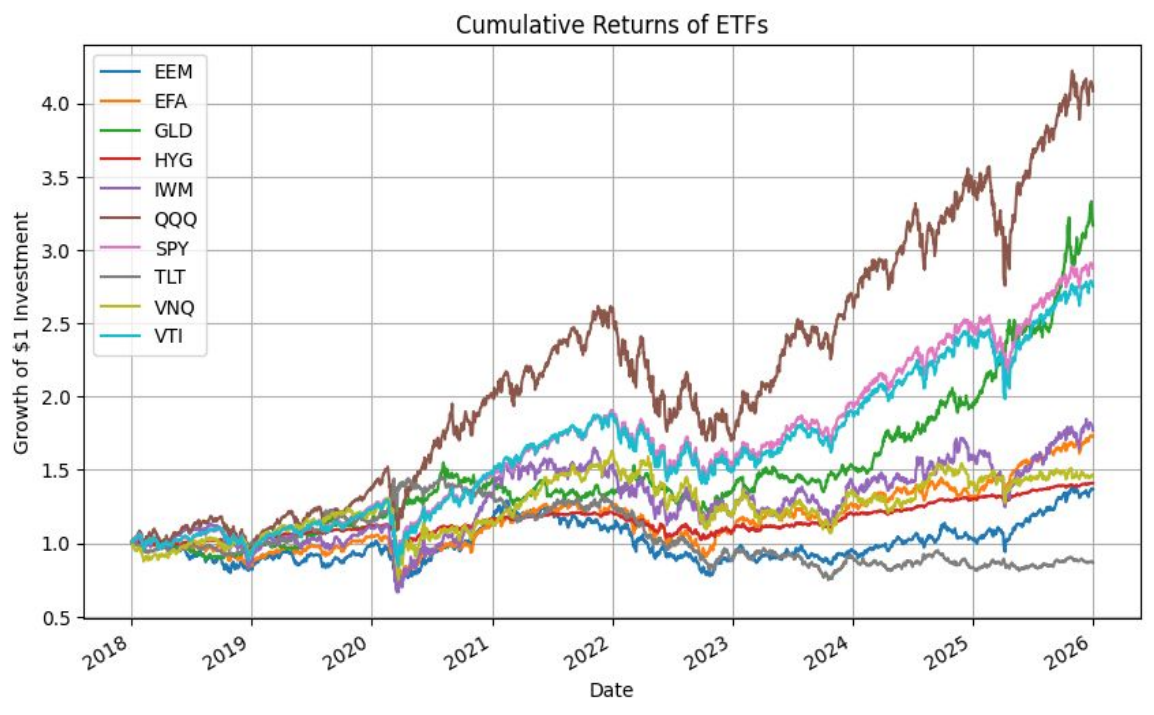
Step 04
Monte Carlo Simulation

Step 05
Mathematical Optimization

Step 06
Practical Optimization

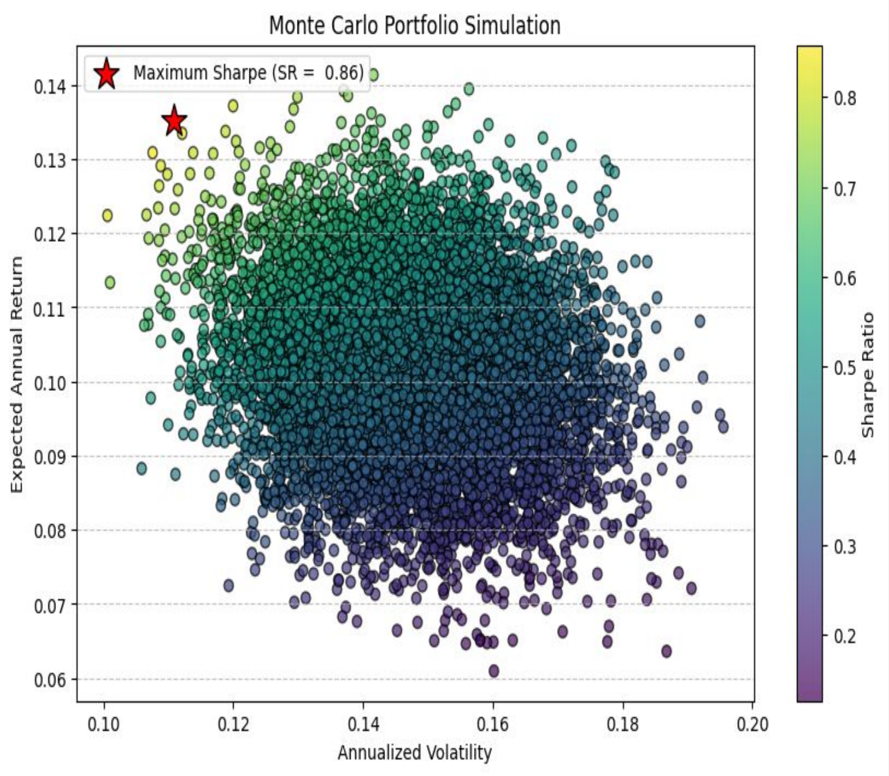
Step 07
Performance Comparison

Data Exploration



This chart illustrates the historical cumulative performance of the selected ETFs over the analysis period. It provides an initial view of return trends before portfolio optimization is applied.

Monte Carlo Simulation



What is Monte Carlo Simulation?
Monte Carlo Simulation is a computational technique that generates thousands of randomly weighted portfolios to evaluate different investment outcomes.

- Generated approximately 10,000 random portfolios.
- Calculated expected return, volatility, and Sharpe Ratio.
- Identified the portfolio with the highest Sharpe Ratio.

Mathematical Optimization

Process:

- Used SciPy's optimization algorithm
- Maximized the Sharpe Ratio
- Identified the mathematical optimal portfolio allocation

Mathematical Portfolio Allocation (unconstrained)

ETF	Weight
SPY	78.04%
VTI	21.96%
Total	100.00%

Key Takeaway: Although this portfolio achieved the highest Sharpe Ratio, it concentrated investments in only two ETFs.

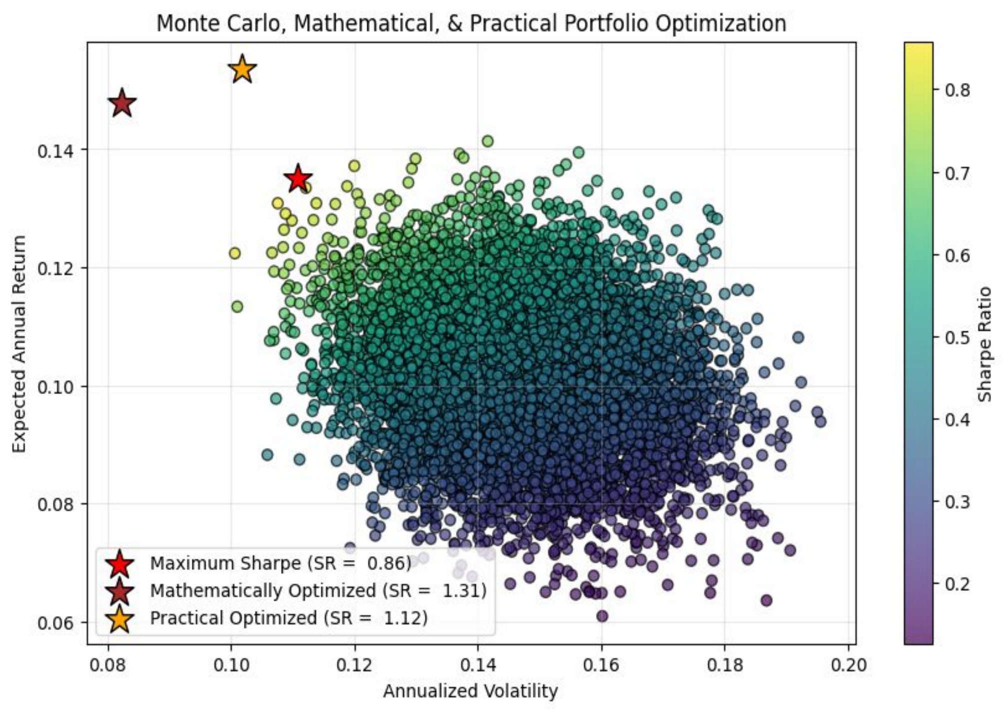
Practical Optimization

Added:

- Maximum 30% allocation per ETF

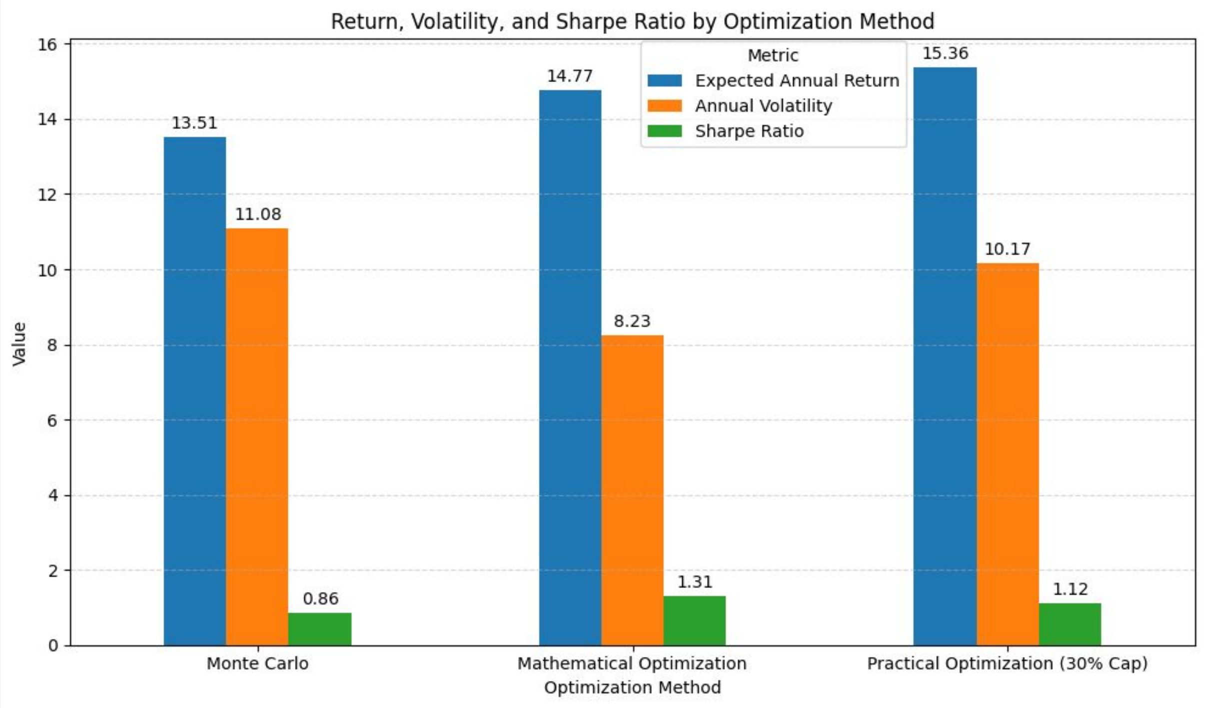
Purpose:

- Improve diversification
- Reflect real-world investing



Key Takeaway: A 30% allocation cap produces a more diversified portfolio while maintaining strong risk-adjusted performance.

Results Comparison



Mathematical optimization achieved the highest Sharpe Ratio, while practical optimization maintained similar performance with substantially better diversification.

Recommended Portfolio Allocation

Constrained Optimization (30% Cap)

Applying a 30% allocation cap per ETF reduces concentration risk while maintaining strong risk-adjusted returns. The result is a more diversified and practical investment portfolio.

Final Portfolio Allocation (30% Cap)

ETF	Weight
SPY	**30.00%**
VTI	**30.00%**
QQQ	**27.52%**
EEM	**12.48%**
Total	**100.00%**

Key Takeaway: A 30% allocation cap reduces concentration risk while preserving strong portfolio performance.

Recommendation Rationale

The constrained optimization portfolio is recommended because it maintains strong risk-adjusted performance while improving diversification.

Constrained optimization portfolio provides:

1. Better risk-return balance
2. Reduced concentration risk
3. More realistic for implementation
4. Improved diversification

Limitations & Future Improvements

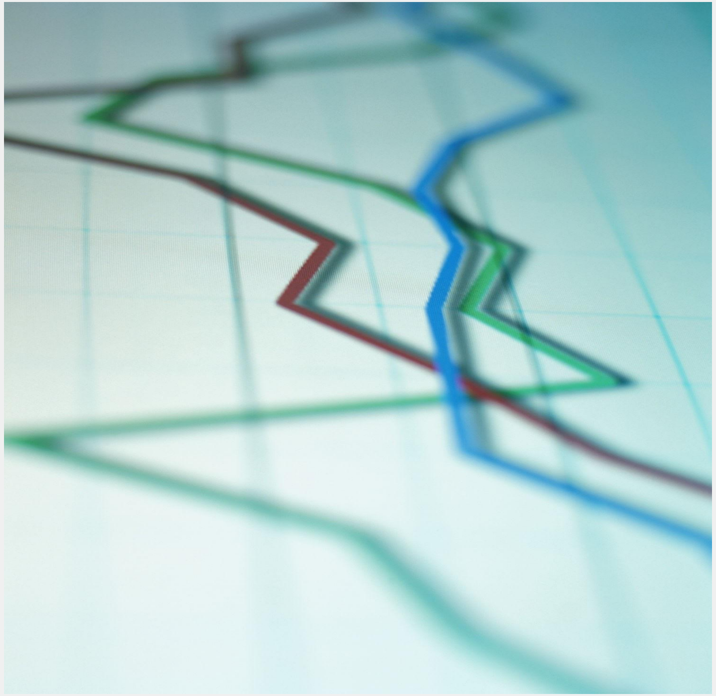
Limitations

- Only historical data
- No transaction costs
- No taxes
- Market conditions change

Future Improvements

- Additional ETFs
- More risk metrics
- Rolling-window analysis
- Interactive dashboard

Key Takeaways



Historical ETF data can support portfolio optimization.

Different optimization methods produce different allocation strategies.

Maximizing the Sharpe Ratio alone may lead to concentrated portfolios.

Practical constraints can improve diversification while maintaining competitive performance.

Thank you!

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